



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 3	
Name of Student	Raghav Dhoot
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Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Simulating Git Collaboration & Conflict Resolution.
Implementation & Theory	<pre>(base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git checkout main git pull origin main git status Already on 'main' Your branch is up to date with 'origin/main'. From github.com:dhootraghav/devops-coursework-2026 * branch main -> FETCH_HEAD Already up to date. On branch main Your branch is up to date with 'origin/main'. nothing to commit, working tree clean</pre> git checkout main + git status — Confirmed starting on main, clean relative to origin/main before creating the conflicting branch. <pre>(base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git checkout -b conflicting-feature Switched to a new branch 'conflicting-feature'</pre> git checkout -b conflicting-feature — Created a new branch conflicting-feature from the current state of main, so both branches would diverge from the same point. <pre>(base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % nano README.md (base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git add README.md git commit -m "Update README first line on feature branch" [conflicting-feature 6b18db2] Update README first line on feature branch Committer: Raghav Dhoot <raghavdhoot@Raghavs-MacBook-Air-2.local></pre> nano + git add + git commit (feature branch) — Modified the first line of README.md on conflicting-feature and committed the change.

```

((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % nano README.md
((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git add README.md
git commit -m "Update README first line on main"
[main 1f78be1] Update README first line on main
Committer: Raghav Dhoot <raghavdhoot@Raghavs-MacBook-Air-2.local>

```

git checkout main + nano + git add + git commit (main) — Switched back to main, modified the same line differently, and committed — setting up the conflicting histories.

```

((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git merge conflicting-feature
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

```

git merge conflicting-feature — Attempted the merge; Git reported CONFLICT (content): Merge conflict in README.md since both branches changed the same line.

```

((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

```

git status (unmerged paths) — Confirmed the conflict state, showing README.md under "Unmerged paths."

```

((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % nano README.md
((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % cat README.md
# DevOps Coursework - Merged Update (Main + Feature)
This repository contains my DevOps Lab assignments.

```

nano + cat README.md (resolved) — Manually edited the file to remove the conflict markers (<<<<<<, =====, >>>>>>) and settled on a final merged version of the line.

```

((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git add README.md
git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

It took 12.32 seconds to compute the branch ahead/behind values.
You can use '--no-ahead-behind' to avoid this.

All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   README.md

```

git add + git status (all conflicts fixed) — Staged the resolved file; status confirms "All conflicts fixed but you are still merging."

```

((base) raghavdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git commit
[main da3e172] Merge branch 'conflicting-feature'
Committer: Raghav Dhoot <raghavdhoot@Raghavs-MacBook-Air-2.local>

```

git commit -m "Merge conflicting-feature..." — Completed the merge with a commit that ties both branch histories together.

```
((base) raghvdhoot@Raghavs-MacBook-Air-2 devops-coursework-2026 % git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 936 bytes | 936.00 KiB/s, done.
Total 9 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 2 local objects.
To github.com:dhoottraghav/devops-coursework-2026.git
 6cf9859...da3e172  main -> main
```

git push origin main — Pushed the resolved merge commit to GitHub.